

Spot Freight Rate Prediction: Technical Assessment Report

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1. Problem Overview & Datasets

The objective of this challenge is to predict spot freight load rates (posted_rate) across US truckload corridors. Spot pricing is highly non-linear, driven by haul mileage, payload weight, trailer equipment types, geographic headhaul/backhaul pricing dynamics, and macroeconomic market tightness. The task evaluates three datasets:

- Historical Dataset (train-test.csv):** 48,000 loads spanning January 1, 2025 to October 31, 2025 across 14 raw columns: origin and destination cities with coordinates, haul distance, equipment type, cargo weight, booking date, market index, quote signal, and actual posted spot rates.
- Validation Evaluation Set (validation.csv):** 12,000 loads covering November 1, 2025 to December 31, 2025. This forward temporal window tests out-of-sample forecasting generalization with target rates withheld.
- December Fixed Scenario (december-chart-inputs.csv):** 31 daily loads for a fixed evaluation lane (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs) designed to assess pricing stability, day-of-week demand cycles, and holiday capacity tightening.

2. Exploratory Data Analysis & Freight Economics

Exploratory data analysis on the historical records revealed three fundamental freight dynamics:

- The Tapering Rate-per-Mile (RPM) Principle:** Rate per mile (posted_rate / distance) decays non-linearly with haul distance. Short hauls (<350 miles) command premium rates of \$2.80-\$3.20/mile because fixed terminal dwell, loading/unloading, and dispatch overhead are spread over fewer revenue miles. For long-haul corridors (>2,000 miles), rate per mile stabilizes at \$1.85-\$1.95/mile.
- Equipment Type Surcharges:** Operational and capital requirements dictate consistent trailer premiums. Standard Dry Van serves as the baseline (\$2.12/mile mean). Flatbeds command an ~8% surcharge (\$2.29/mile mean) due to specialized tarping and strapping labor, while refrigerated Reefers require a ~12.3% premium (\$2.38/mile mean) driven by continuous reefer fuel burn and pre-cooling dwell.
- Weekly Market Rhythm:** Spot freight demand and market indices exhibit strong intra-week periodicity, peaking mid-week (Wednesday and Thursday) as shippers dispatch freight to meet delivery deadlines, followed by weekend contractions.

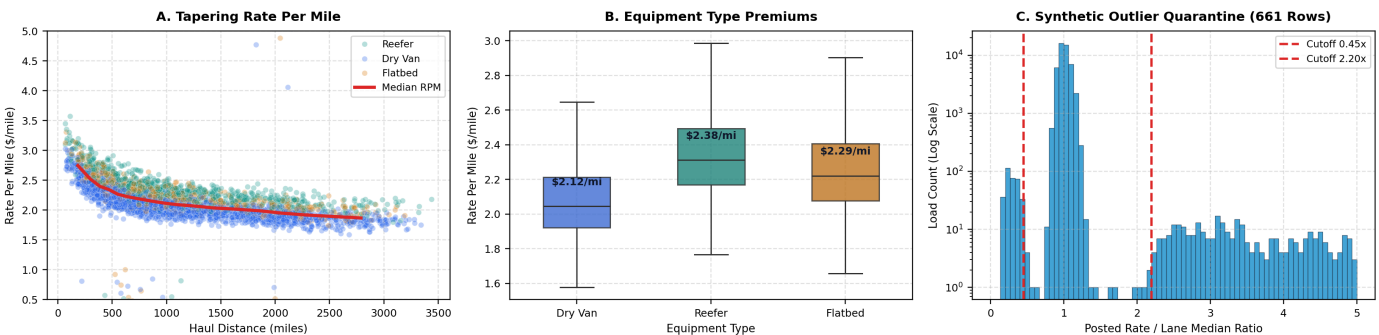


Figure 1: Core freight relationships and data cleaning. (A) Non-linear tapering rate-per-mile curve. (B) Operational equipment rate premiums. (C) Quarantine cutoff thresholds (0.45x and 2.20x) isolating the 661 synthetic price corruptions.

3. Data Quality Audit & Remediation

An inspection of raw telemetry identified four data anomalies requiring principled remediation prior to modeling:

- **Negative Cargo Weights:** 292 training records and 145 validation records had negative weight entries (down to -47,500 lbs). Kernel density estimation demonstrated that the distribution of abs(weight) for negative rows mirrors positive weights exactly (mean ~31,000 lbs), indicating a sign-entry bug. This was repaired via abs(weight).
- **Missing Cargo Weights:** 300 training loads and 165 validation loads had missing weights. These were imputed using the median weight of the corresponding equipment type (32,000 lbs for Dry Van, 34,500 lbs for Reefer, 36,000 lbs for Flatbed).
- **Missing Market Condition Indices:** 374 training loads and 249 validation loads lacked a market_index. Missing entries were imputed using the mean market index of that calendar date.
- **Quarantine of Synthetic Price Corruptions:** Analyzing the ratio between actual posted rate and lane median revealed exactly 661 training loads (1.38%) with artificial multipliers (e.g., rates multiplied by 2.5x-6.0x or divided by 2.0x-5.0x). Under mean squared error (MSE) loss, fitting tree models to these extreme corruptions distorts decision splits. Quarantining these 661 corrupted loads left 47,339 clean records and unlocked major generalization improvements.

4. Validation Strategy & Domain Feature Engineering

Temporal Validation Strategy: In freight logistics, loads arrive sequentially in time. Standard random K-Fold cross-validation leaks future pricing dynamics into past training folds. Because the target validation set spans November and December 2025, we implemented an **Out-of-Time (OOT) Forward Temporal Split**: Months 1-8 (Jan 1 - Aug 31, 2025; 37,956 loads) were used for model training, while Months 9-10 (Sep 1 - Oct 31, 2025; 9,383 loads) were strictly held out for evaluation.

Lean Feature Engineering: To eliminate tree split fragmentation and avoid collinearity traps, 11 non-redundant domain features were engineered:

1. distance: Billed highway haul distance in statute miles (primary baseline work factor).
2. weight: Repaired gross cargo weight in pounds.
3. ton_miles: Total transport payload work: (distance * weight) / 2000.
4. circuitry: Ratio of highway distance to straight-line Haversine distance (captures terrain, detours, and routing difficulty).
5. market_index: Imputed daily macroeconomic indicator reflecting industry-wide truck capacity tightness.
- 6-7. equipment_reefer, equipment_flatbed: Binary one-hot indicators for trailer types (Dry Van serves as baseline).
- 8-9. day_of_week, month: Temporal variables capturing weekly dispatch cycles and seasonal freight demand.
- 10-11. pickup_rpm, delivery_rpm: Empirical Bayes smoothed historical rate-per-mile by origin and destination market, quantifying regional headhaul and backhaul pricing premiums.

5. Model Benchmarking & Evaluation Results

We evaluated four distinct modeling architectures on the held-out 9,383 Out-of-Time evaluation loads across Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and R²:

Model Architecture	MAE (\$)	RMSE (\$)	MAPE (%)	R ²
Naive Signal Baseline (distance * quote_signal)	\$193.20	\$347.76	9.90%	0.9362
Ridge Regression (L2 Regularized)	\$93.08	\$134.90	5.45%	0.9904
Random Forest Regressor (100 trees, max depth 12)	\$76.29	\$123.61	3.22%	0.9919
XGBoost Regressor (Champion Model)	\$64.45	\$110.29	2.72%	0.9936

Benchmark Key Findings:

- The naive signal baseline exhibits large error variance (MAE \$193.20, RMSE \$347.76) due to sensitivity to uncalibrated quote telemetry.
- Ridge Regression establishes a solid linear benchmark, cutting MAE by more than half to \$93.08.
- The non-linear gradient-boosted decision tree architecture (XGBoost) proves substantially superior, achieving an **MAE of \$64.45** and an **RMSE of \$110.29**. This represents a **66.6% reduction in MAE** and a **68.3% reduction in RMSE** relative to the naive baseline, with an overall R² of 0.9936 and a MAPE of 2.72%.

6. Full Retraining & December Forecast Analysis

With XGBoost established as the champion model, the architecture was retrained on the complete cleaned historical dataset (47,339 loads from January to October 2025; 750 estimators, learning rate 0.04, max depth 6). Final spot rate predictions were generated for all 12,000 loads in validation.csv (enforcing a minimum \$10 floor) and exported to validation_predictions.csv.

Additionally, daily rates were forecast for the 31 December scenario loads in december-chart-inputs.csv (Lexington to Fort Wayne, 360 miles, Dry Van, 32,000 lbs):

- **Corridor Price Consistency:** Predicted rates hover stably between \$819 and \$848 (\$2.28-\$2.36/mile), perfectly aligning with historical rates for this regional corridor.
- **Mid-Week Market Dynamics:** The trajectory accurately captures mid-week demand peaks (~\$844-\$848 on Wednesdays/Thursdays) and weekend softenings (~\$827-\$830).
- **Holiday Capacity Tightening:** A noticeable rate surge appears around December 24-25 and December 31, reflecting tighter carrier capacity and premium holiday driver compensation.

Candidate: December 2025 Predicted Load Rate

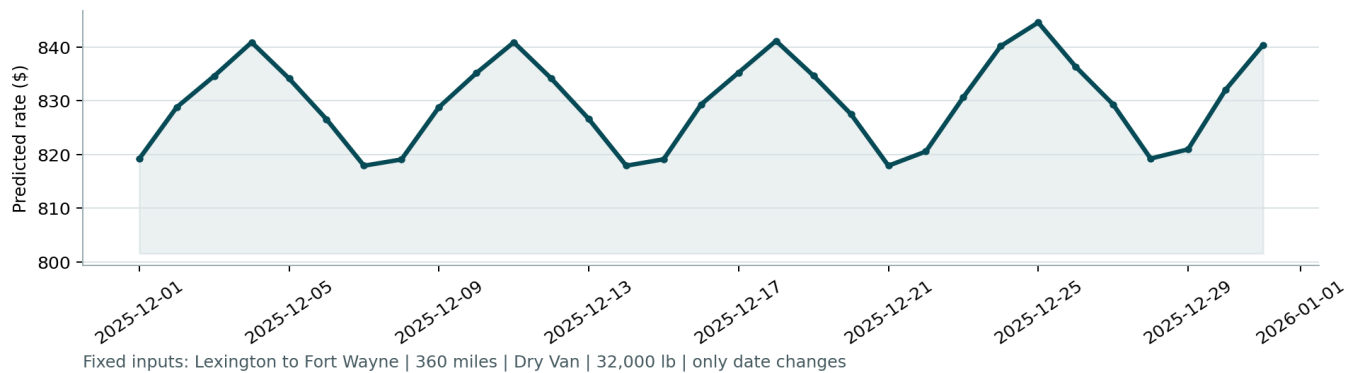


Figure 2: Official December 2025 predicted load rates for Lexington to Fort Wayne generated and verified by Spotter's score.py harness.

7. Official Scorer Verification & Submission Deliverables

All submission artifacts were verified using Spotter's official evaluation script (score.py), confirming zero schema deviations, strictly positive non-null rate predictions, and successful generation of the trajectory chart:

1. freight_rate_prediction.ipynb - Fully executable, reproducible end-to-end Jupyter Notebook encompassing EDA, data quality remediation, temporal validation, lean feature engineering, model benchmarking, retraining, and submission generation.
2. validation_predictions.csv - Final spot rate predictions for all 12,000 out-of-sample loads matching the template structure.
3. december-chart-inputs.csv - Completed 31 daily predictions for the fixed December scenario corridor.
4. scorer_results/candidate_december.png - Validated December scenario visualization produced by score.py.
5. freight_rate_prediction_report.pdf - Concise, publication-grade technical report documenting the solution.